136 GROUND WATERS OF WESTERN SAN Ml<36 COtfFTY, CAUf.

middle part loss and gain balance each, other, and the water level remains stationary from about October 1 to the first- big storm pro-ducing flood run-off. Thus the observed fluctuation are the effects of gain and loss and the tendency of the water surface to assume a horizontal position in each ground-water reservoir.
In the fourth group the continued flow of the stream replenishes the ground-water supply at the head of the basin throughout the year and thus counteracts the tendency of the water table to fall. The rate of evaporation begins to decrease in September (Tables 25, 26, and 27, pp. 101, 102), at which time the water table begins to rise through replenishment by lateral percolation f^om the near-by
stream.

TABLE 34. Duration of flow of principal streams in San Diego C&i'.nfy at typical gaging stations.

San Luis Rey River near Pala.

Year (July 1 to June 30).

1903-4..................
1904-5..................
1905-6..................
1906-7..................
1907-8..................
1908-9..................
1909-10.................
1910-11.................
1911-12.................
1912-13.................
1913-14.................
1914-15.................

Corrected aver-

Date that
continuous
flow com-
menced.

July 1

...do.......
...do.......
...do.......
...do.......
...do.......
...do.......
...do.......
...do.......
...do.......
...do.......

Date that
flood flow
com-

meneed.a

Mar. 23

Jan. 19
Nov. 23
Oct. 16
Jan. 13
Nov. 11

Date of
last flood
flow."

Stay 26
(June 12)

May 4
May 2

Apr. 19
May 16

May 25

Date that
continuous
flow
ceased.

...do.......
...do.......
...do.......
...do.......
...do.......
...do.......
...do.......
...do.......
...do.......
...do.......
...do.......

Number

of days df con-
tinuous
flOYT.

365
365
r«s
365

365 "65
f«5
365
365
365
365

365

365

Number

of days

of flood flow.<t

61
flfwl^

143
209
144

135
159
106
74
37
118

160

125

100

Minimum
flow of
stream

during
nonflood

period.

Sec. ft.
1.0
2.0
2.5
3.0
0.6
2 A
5.0

(4.0)

3.0
o 0

1.0

San Luis Rey River at Bonsall.

1911-12.................

1913-14.................
1914-15................

Corrected aver-

Tlor* 1 9
Tjprt -I
Nov. 15

(Mar. 6)
Jan. 10
Jan. 18

May 16
Mar. 30
May 8

177
207
997

167)
51
107
160

96

Oft

0
0
0
0

a Flood flow considered as flow in excess of 14 second feet.

6 Observed average multiplied by ratio of observed 17-year average on San Diego River at diverting dam to observed average for period 1903-4 to 1914-15.